Department of Computer Engineering
Faculty of Engineering and Technology
Jamia Millia Islamia, New Delhi

Academic Session 2025-2026

2nd Mid Semester Examination B.Tech. (Computer Science & Engineering) Data Science 4th Sem

SUBJECT: DSC-401 **Data Mining** **Credits =3**

TIME: 01Hours **Max. Marks: 12**

Note:

1. *M-Marks, L-BloomsLevel, C–CourseOutcome*
2. *Electronic gadgets are not allowed in the examination hall like mobile phone, smart watch etc.*
3. *Students will only be allowed to use non-programmable scientific calculator.*
4. **All questions are compulsory**

<table>
<tr><th>S.No.</th><th>Questions</th><th>M</th><th>L</th><th>C</th></tr>
<tr><td>1.</td><td>(i) Explain simple linear regression with the help of suitable graphical representations.
(ii) Derive the Mean Squared Loss Function for the simple linear regression function
(iii) What is the Mean Squared Error for the following dataset
<table>
<tr><td>X</td><td>2</td><td>3</td><td>5</td><td>6</td><td>8</td></tr>
<tr><td>Y-true</td><td>6</td><td>4</td><td>5</td><td>8</td><td>10</td></tr>
<tr><td>Y-Pred</td><td>7</td><td>5</td><td>6</td><td>7</td><td>9</td></tr>
</table></td><td>6</td><td>3</td><td>3</td></tr>
<tr><td>2.</td><td>Consider the following dataset of students' study habits and exam results where each student passed or failed the exam based on their study hours and attendance.
<table>
<tr><th>Student</th><th>Study Hours (H)</th><th>Attendance (A)</th><th>Class (Y)</th></tr>
<tr><td>S1</td><td>High</td><td>Good</td><td>Pass</td></tr>
<tr><td>S2</td><td>Medium</td><td>Average</td><td>Pass</td></tr>
<tr><td>S3</td><td>Low</td><td>Good</td><td>Fail</td></tr>
<tr><td>S4</td><td>Medium</td><td>Poor</td><td>Pass</td></tr>
<tr><td>S5</td><td>Low</td><td>Average</td><td>Fail</td></tr>
<tr><td>S6</td><td>Medium</td><td>Poor</td><td>Fail</td></tr>
<tr><td>S7</td><td>High</td><td>Average</td><td>Pass</td></tr>
<tr><td>S8</td><td>Low</td><td>Poor</td><td>Fail</td></tr>
<tr><td>S9</td><td>High</td><td>Good</td><td>Pass</td></tr>
<tr><td>S10</td><td>Medium</td><td>Average</td><td>Pass</td></tr>
<tr><td>S11</td><td>Medium</td><td>Good</td><td>?</td></tr>
<tr><td>S12</td><td>High</td><td>Poor</td><td>?</td></tr>
</table>
(i) Computer the prior probabilities.
(ii) Compute the all the conditional probabilities for each feature given each class.
(iii) Using the Naïve Bayes classification, predict the class for a new student S11 and S12.</td><td>6</td><td>4</td><td>4</td></tr>
</table>

Bloom's Taxonomy Levels (1-Remembering, 2-Understanding, 3- Applying, 4- Analyzing, 5- Evaluating, 6- Creating)